MRI signs helpful in the differentiation of patients with anterior ischaemic optic neuropathy and optic neuritis

Valentina Daphne Petroulia,[1] Dominik Brügger ,[2] Robert Hoepner,[3] Rino Vicini,[2] Anna Winklehner,[4] Mathias Abegg ,[5] Franca Wagner [4]

[1]Diagnostic and Interventional Neuroradiology, Inselspital Universitatsspital Bern, Bern, Switzerland
[2]Ophthalmology, Inselspital University Hospital Bern, Bern, Switzerland
[3]Neurology, Inselspital University Hospital Bern, Bern, Switzerland
[4]Diagnostic and Interventional Neuroradiology, Inselspital University Hospital Bern, Bern, Switzerland
[5]Ophthalmology, University of Bern, Bern, Switzerland

Correspondence to
Dr Franca Wagner, Diagnostic and Interventional Neuroradiology, Inselspital University Hospital Bern, Bern 3010, Switzerland; franca.wagner@insel.ch

VDP and DB contributed equally.

Received 26 April 2021
Accepted 1 July 2021
Published Online First 19 July 2021

To cite: Petroulia VD, Brügger D, Hoepner R, *et al*. *Br J Ophthalmol* 2023;**107**:121–126.

ABSTRACT

Background/Aims The aim of this study was to identify specific MRI characteristics of anterior ischaemic optic neuropathy (AION) and optic neuritis (ON) that would aid in the differentiation between these two diagnoses.

Methods We retrospectively analysed a consecutive case series including all patients with an MRI study of brain and orbit and the clinical diagnosis of either ON or AION. We examined the scans for restricted diffusion of the optic nerve, optic sheath diameter, enhancement and location of enhancement of the optic nerve and distribution of the white matter lesions.

Results Fifty patients met the inclusion criteria. We found an accuracy of 0.98 for the discrimination between AION and ON based solely on parameters extracted from MRI data. Dominance analysis to determine the most influential parameters showed that the enhancement pattern of the optic nerve and distribution of the white matter lesions had the biggest impact on the classification and led to a discrimination accuracy of 0.9 when used alone.

Conclusion In patients with an inconclusive clinical diagnosis, optic nerve enhancement pattern and distribution of white matter lesions can aid in the diagnosis and differentiation between AION and ON. Diffusion-weighted imaging did not add significant information to the diagnosis or help to differentiate between the two conditions.

INTRODUCTION

Anterior ischaemic optic neuropathy (AION) is one of the most prevalent and debilitating optic neuropathies in the middle-aged and older population.[1] It is an acute disease that is presumed to be caused by a circulatory insufficiency or an interruption of the blood flow to the optic nerve leading to different degrees of vision loss.[2] In line with the presumed pathomechanism, the main risk factors for AION are vascular, including age, arterial hypertension, sleep apnoea and giant cell arteritis. The recognition and treatment of the last of these is of particular interest owing to a high risk of fellow eye involvement.[1] The common optic neuropathy in the younger population is inflammatory optic neuritis (ON), which is often the first sign of multiple sclerosis (MS).

Clinically, ON is associated with reduced visual acuity, impaired colour vision and a central scotoma in the visual field. AION is associated with disc swelling in the acute phase and nerve fibre bundle defects in the visual field, which may or may not reduce visual acuity. Age, visual field, presence of pain and appearance of the optic disc usually allows the two entities to be readily distinguished in the acute phase. The distinction may not always be straightforward, however, as both entities may cause pain and disc swelling as well as affecting visual acuity. The final diagnosis may only be made retrospectively based on the time course ending with good recovery in cases of ON and little or no recovery for AION. Early differentiation between the two entities, therefore, is critical for effective management, treatment and follow-up.[3 4]

In cases of inconclusive clinical assessment, an MRI is often acquired. The primary role of MRI is to exclude pathologies such as demyelination (MS) or mass lesions that may compress the optic nerve. At the level of the optic nerve, ischaemia and neuritis are usually not differentiated in neuroimaging. Since the optic nerve, like the spinal cord, is an 'extension' of the central nervous system (CNS), the pathophysiology should also be analogous. Thus, acute ischaemia of the optic nerve is comparable to that of the CNS, with similar imaging characteristics. These include hyperintensity in the diffusion-weighted imaging (DWI) with correlated hypointensity in apparent diffusion coefficient (ADC) owing to cytotoxic oedema and consequent restriction of water diffusion.[5 6] The pathophysiology causing the diffusion restriction in the acute setting of ON though is different to that of AION. In ON, diffusion restriction is due to the inflammation of the optic nerve and the consequent accumulation of macrophages, astrocytes and lymphocytes, as well as oedema of the optic nerve,[6] causing a restriction of diffusion of the water molecules of the optic nerve and an associated decrease of the ADC values. By contrast, the potential decrease of ADC values in the optic nerve head in non-arteritic-AION (N-AION) is explained by the decrease of blood flow or occlusion of the short posterior ciliary arteries. Hence, despite the different pathomechanisms leading to diffusion restriction of the optic nerve head and optic nerve, ultimately, differentiation between ON and N-AION on the basis of DWI is not expected to be possible.[7]

Even though MRI is routinely performed for ON and is common in cases of atypical AION, MRI criteria for differentiating one from the other have not been described in the medical literature.

The purpose of this study was to evaluate the role of MRI in the assessment of ischaemic optic nerve damage in anterior ischaemic optic neuropathies and to identify specific imaging characteristics of cerebral and orbital MRI and postcontrast enhancement. To our knowledge, this is the first study focusing on the role of MRI-based biomarkers for the differentiation between AION and inflammatory ON, irrespective of any association of these clinical diagnoses with age or sex.

MATERIALS AND METHODS

Subjects

We retrospectively reviewed our MRI database to find all patients who had presented at our emergency department between March 2010 and December 2018 with new visual disturbances with unknown aetiology as leading symptom and who had undergone an emergency MRI scan of the brain and/or dedicated orbital sequences. We then selected all patients for whom concurrent ophthalmic examination data were accessible. From these patients, we selected all those that had a clinically confirmed diagnosis of either AION or ON. Diagnosis was then revisited by experienced neuro-ophthalmologists (RV, MA) to ensure that only cases with a clear diagnosis were included.

Inclusion criteria were: (a) age over 18 years, (b) presentation with visual disturbances, (c) complete physical, ophthalmological examination and history available and (d) MRI neuroimaging (brain and/or dedicated orbital MRI scan). Exclusion criteria were: (a) age under 18 years, (b) no informed consent, (c) insufficient or incomplete ophthalmological examination and (d) no or incomplete MRI. Scans with motion artefacts that limited the image quality and read-out were excluded from the outset.

All patients gave written informed consent for the usage of their data for research purposes.

Table 1 Parameters evaluated*

	AION (n=28)	ON (n=22)	P value
Age			
Minimum	45.21	15.86	
Maximum	79.64	67.00	
Mean (SD)	62.47±9.99	41.34±14.05	<0.001
Sex			
Female	11/28 (39.3%)	17/22 (77.3%)	
Male	17/28 (60.7%)	5/22 (22.7%)	0.01
Distribution	Distribution of cerebral T2-hyperintensities (non-specific, microangiopathic, demyelinating)		
Demyelinating	2/28 (7.1%)	10/22 (45.5%)	
Microangiopathic	11/28 (39.3%)	0/22 (0.0%)	
Unspecified	15/28 (53.6%)	12/22 (54.5%)	<0.001
Bright spot	Enhancement of the optic nerve head		
Absent	11/28 (39.3%)	17/22 (77.3%)	
Present	17/28 (60.7%)	5/22 (22.7%)	0.01
Enhancement	Enhancement of the optic nerve		
Absent	23/28 (82.1%)	5/22 (22.7%)	
Present	5/28 (17.9%)	17/22 (77.3%)	<0.001
DWI intraorbital	Diffusion restriction of the optic nerve—intraorbital		
Absent	28/28 (100.0%)	20/22 (90.9%)	
Present	0/28 (0.0%)	2/22 (9.1%)	0.19
DWI intraconal	Diffusion restriction of the optic nerve—intraconal		
Absent	8/28 (28.6%)	10/22 (45.4%)	
Present	20/28 (71.4%)	12/22 (54.5%)	0.25
DWI prechiasmal	Diffusion restriction of the optic nerve—prechiasmal		
Absent	22/28 (78.6%)	21/22 (95.5%)	
Present	6/28 (21.4%)	1/22 (4.5%)	0.12
Optic nerve sheath	Expansion of the optic nerve sheath		
Absent	20/28 (71.4%)	12/22 (54.5%)	
Present	8/28 (28.6%)	10/22 (45.5%)	0.25
Perineural fat tissue	Enhancement of retrobulbar intraconal fat		
Absent	6/28 (21.4%)	5/22 (22.7%)	
Present	22/28 (78.6%)	17/22 (77.3%)	1
Enhancement of extraconal fat	Enhancement of extraconal fat		
Absent	28/28 (100.0%)	22/22 (100.0%)	

*All parameters except the grouping variable 'Diagnosis' represent the assessment of the MRI data by two neuroradiologists.
AION, anterior ischaemic optic neuropathy; DWI, diffusion-weighted imaging; ON, optic neuritis.

Image acquisition and interpretation

Image acquisition was performed either at the University Hospital of Bern or by external referral facilities in the Canton of Bern. The ophthalmological examination took place at the Department of Ophthalmology at the University Hospital of Bern.

All MRI examinations included DWI, T1-weighted and T2-weighted sequences, fluid-attenuated recovery sequences of the brain parenchyma, susceptibility-weighted imaging and contrast-enhanced T1-weighted sequences. Dedicated sequences of the orbit were T2 fat-saturated in the coronal plane and contrast-enhanced T1 fat-saturated in the coronal plane. Two experienced neuroradiologists blinded to the clinical diagnosis independently reviewed the MRI examinations and consensus was sought in cases of disagreement.

Images were qualitatively and quantitatively examined and assessed for restricted diffusion of the optic nerve using high-resolution readout-segmented multishot echo-planar-imaging (RESOLVE, Siemens Healthcare; TR/TE 4850/75; b=1000; field of view 220×220; slice thickness 4 mm) in the intraocular, intraorbital and prechiasmal segment. To avoid orbital motion artefacts due to eye movements during the MR scans, patients were routinely informed before the MRI of the need to focus their gaze on a fixed red point on the scanners and to avoid eye movements during image acquisition. Additionally, before undergoing the MR examination, all patients are required to remove eye make-up and jewellery from piercings, which would cause susceptibility artefacts.

Enhancement was evaluated for every segment of the optic nerve, the optic nerve head (the so-called 'central bright spot') and the perineural fat tissue intraorbitally. The optic nerve sheath diameter, which is known to be independent of sex and body mass index, was measured according to Kim *et al*.[8] These authors stated a normal range of the optic nerve sheath diameter between 3.75 and 6.20 mm with a pathological threshold value above 6.2 mm when measured exactly 3 mm behind the globe. We rated the optic nerve sheath diameter by applying that cut-off for all patients with a binary—yes versus no—differentiation in side-by-side comparison.

White matter lesions were categorised as demyelinating if they had a periventricular, callososeptal, juxtacortical or infratentorial distribution. Microangiopathic lesions were defined as hyperintensities of vascular origin located in subcortical or periventricular white matter.[9] White matter hyperintensities rated 'unspecified' did not meet the criteria for either of the above-mentioned categories. We also recorded the presence or

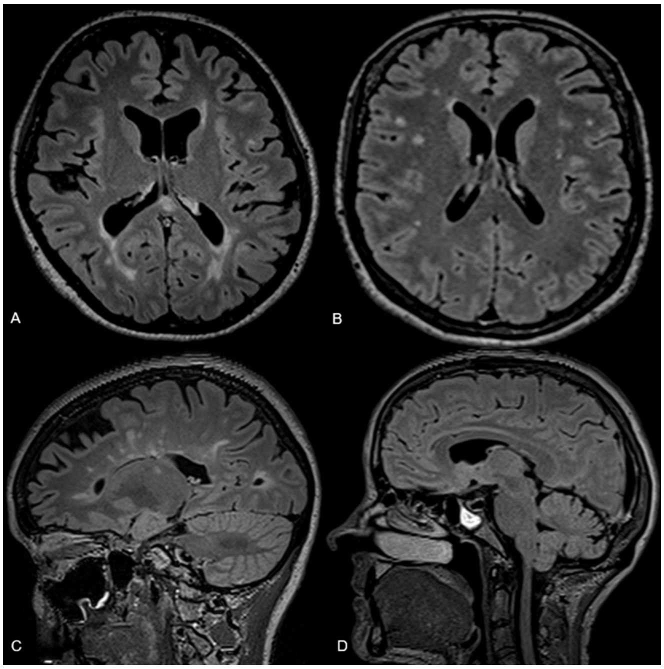

Figure 1 Axial (A, B) and sagittal (C, D) three-dimensional T2-space dark-fluid demonstrates the microangiopathic distribution pattern of white matter lesions in a patient with non-arteritic anterior ischaemic optic neuropathy (N-AION) (B, D) and the typical demyelinating lesions in a patient with optic neuropathy (A, C). Incidental finding of sphenoid sinusitis (left) in the patient with N-AION (D).

absence of an intracranial haemorrhage. Table 1 gives an overview of the parameters and a short description of each.

All images were interpretable for all the above-mentioned criteria by both neuroradiologists. Orbital movement artefacts were present in some scans, but in all scans the optic nerve was assessable in its entirety.

Statistical analysis

To reduce the parameter space, we created a side-invariant version of the data by mirroring the affected side such that all patients had the lesion on the same side. Interaction between the MRI markers in our dataset (all parameters as presented in table 1, except age and sex) was studied by fitting a generalised linear model with binomial distribution and a logit link-function using R and the MASS package.[10] We used automatic parameter reduction based on Akaike information criterion to reduce model complexity.

To determine the importance of all parameters in the prediction, while taking into account potential dependencies between them, we performed dominance analysis[11] using the McFadden index[12] with the dominance analysis package[13] in R.

Unless otherwise stated, we report results as mean±SD.

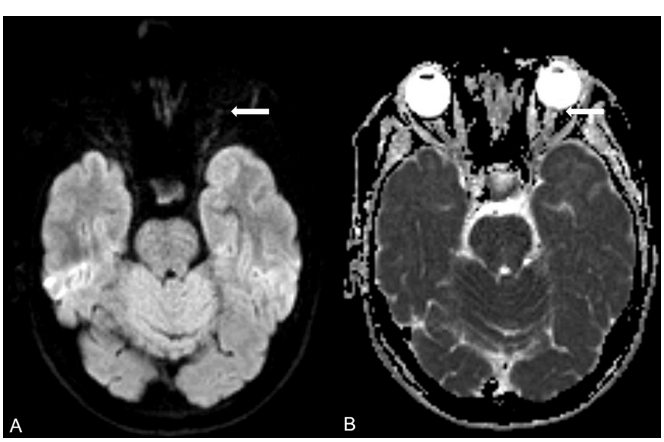

Figure 2 Axial diffusion-weighted imaging (DWI) (A) and apparent diffusion coefficient map (ADC) (B) of a patient with anterior ischaemic optic neuropathy showing high signal intensity in the DWI with low signal intensity in the ADC map; this is indicative of diffusion restriction of the optic disc of the left optic nerve (white arrow).

RESULTS

Fifty patients were included in this study, with a mean age of 53.17±15.87 years; 28 patients were diagnosed with AION, either arteritic or non-arteritic, and 22 with ON.

The time between symptom onset and MRI acquisition varied between 5 hours and 30 days. Patients in the AION group had a mean age of 62.47±9.99 years, whereas the age of those in the ON group was 41.34±14.05 years. This age difference was statistically significant (Welch's t-test $p\leq0.01$). Women were significantly more often affected by ON than by AION (Fisher's exact test $p=0.01$, OR 5.25).

The scoring of the MRI data by the two independent raters resulted in a good agreement for all parameters assessed with a Cohen's kappa of 0.998. We used stepwise model fitting to automatically select predictive variables: starting with full quadratic interaction terms of the parameters described in table 1 excluding age and sex. The resulting model after the parameter reduction step is dependent on the distribution of white matter lesions, the central bright spot, enhancement of optic nerve, DWI optic nerve—prechiasmal segment—and the optic nerve sheath. Application of the resulting generalised linear model led to a total classification accuracy of 0.98. The validity of the model was checked using hold-out validation with 80%/20% sections.

Dominance analysis was performed to assess the contribution of the different parameters to the prediction: enhancement showed the biggest impact on the classification, with a McFadden index of 0.35; distribution showed a similarly high contribution with a value of 0.32. The central bright spot contributed with 0.14, DWI optic nerve—prechiasmal segment— with 0.07, and optic nerve sheath with 0.07.

Modelling with a subset of the available variables using the three most important parameters 'distribution', 'enhancement' and 'central bright spot' led to a classification accuracy of 0.92, and using the two parameters 'distribution' and 'enhancement' alone led to a classification accuracy of 0.90.

We tested the diagnostic value of DWI sequences for discrimination between AION and ON by using only DWI restriction for the modelling, which led to a prediction accuracy of 0.68. By contrast, using all parameters except the DWI led to a model with the same classification accuracy (0.98) as the full model with all parameters. Therefore, the DWI sequence added no information to our evaluation process.

DISCUSSION

We show that neuroimaging, that is, imaging of the orbits, can reliably differentiate AION from ON. The main differences are the distribution of cerebral T2-hyperintensities, presence and localisation of enhancement of the optic nerve and presence of a central bright spot. Up to now, the role of MRI in cases of optic neuropathy has mainly been to rule out underlying pathologies like MS, tumours, etc. The diagnosis itself was solely based on clinical criteria. Our data show that MRI may help in making the correct diagnosis in unclear cases. Since ON usually requires high-dose steroid treatment, the management of AION is fundamentally different. A timely diagnosis of AION and of possibly associated giant cell arteritis may be critical to save the fellow eye.

A study by Rizzo *et al*[14] reported that the abnormal MRI finding in patients with N-AION was an increased signal of the optic nerve in the short-tau inversion recovery sequence. The use of T1-weighted MRI with gadolinium to evaluate ischaemic optic nerve damage has been reported, but did not reveal significant abnormalities in patients with N-AION. A more recent study in a larger group of patients, which took into consideration their age and sex, revealed no significant DWI restriction in patients with N-AION, whereas a significant enhancement of the optic nerve head (central bright spot sign) was observed.[15 16] Many independent studies have reported differences in the imaging findings between ON and ischaemic optic neuropathy (ION).[17–21] However, none of them had a sufficient number of patients or compared the imaging characteristics of ON with the broad spectrum of ischaemic optic neuropathies, but rather compared patients with N-AION.

To our knowledge, this is the first study to present a statistical analysis of MRI characteristics in patients with ON and AION. In our study, the number of patients with cerebral parenchymal lesions, that is, white matter hyperintensities with a microangiopathic distribution like confluent patchy T2-hyperintensities or FLAIR-hyperintensities with mainly periventricular, subcortical or deep white matter distribution, with or without lacunes (figure 1) and AION, indicative of small vessel disease was in line with the findings of Kim *et al* and Arnold.[22 23] These authors found a higher incidence of small vessel disease in patients with N-AION than in controls, explained by the different pathophysiology of the two diseases.[7 24 25] In contrast, Rizzo *et al*[14] postulated that the presence of cerebral white matter lesions was not helpful in differentiating between ON and N-AION.

The expansion of the optic nerve sheath was not significantly different between the ON and AION groups. In the study by Adesina *et al*,[15] most of the patients with N-AION showed postcontrast enhancement and no diffusion restriction of the optic disc (figure 2). In contrast, in the patients with ON, diffusion restriction (figure 3) and postcontrast enhancement of the intraorbital segment of the optic nerve was present, after adjustment for age and sex. In our findings, the presence of enhancement of the optic nerve head (central bright spot sign)[16] (figure 4) differed significantly between the two groups, in keeping with the findings of Adesina *et al*.[15] Relying solely on the MRI findings, that is, presence of positive optic disc findings, was not sensitive enough for a definitive diagnosis of either ON or N-AION. However, in our study, relying on the presence or absence of white matter hyperintensities with microangiopathic distribution, enhancement of the optic nerve and the presence

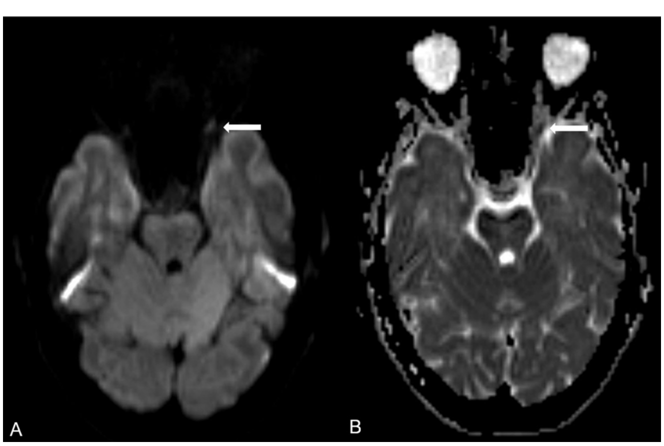

Figure 3 Axial diffusion-weighted imaging (DWI) (A) and apparent diffusion coefficient map (ADC) (B) of a patient with optic neuritis revealing diffusion restriction in the distal intraorbital segment of the left optic nerve (white arrow).

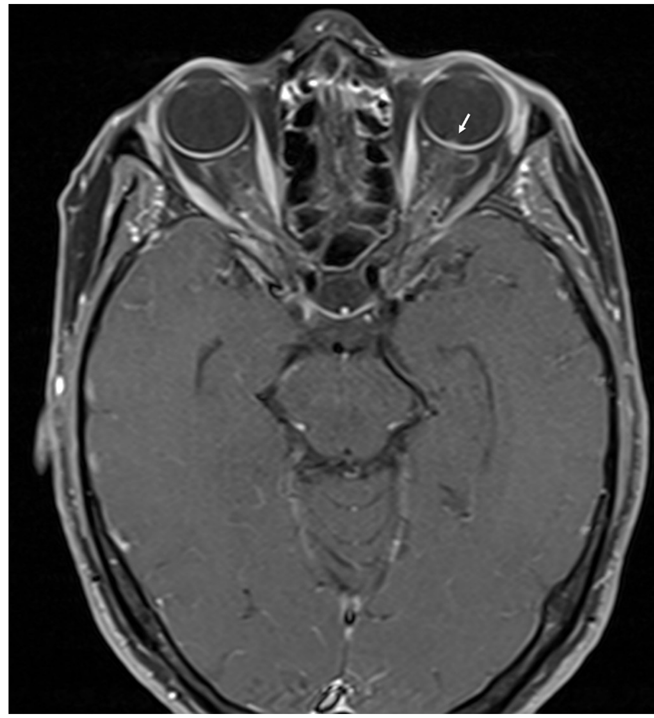

Figure 4 The axial T1-weighted image after contrast with fat saturation in a patient with non-arteritic anterior ischaemic optic neuropathy, left, reveals the optic nerve head enhancement, the so-called ‘central bright spot’ sign on the left side.

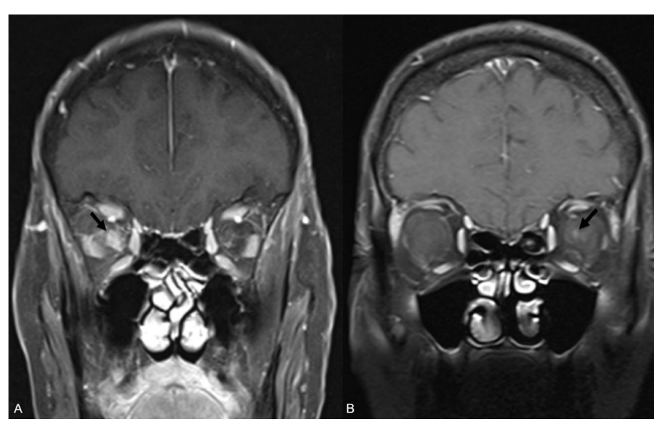

Figure 5 The coronal T1-weighted image after contrast with fat saturation in a patient with optic neuropathy right (A) reveals enhancement of the optic nerve and stranding and enhancement of the perineural fat (black arrow). In comparison, the coronal T1-weighted image with fat saturation after contrast in a patient with non-arteritic anterior ischaemic optic neuropathy (B) shows enhancement of the optic nerve head, also known as the classical ‘central bright spot’ sign (black arrow).

of the central bright spot sign allows for diagnosis and differentiation between ON and AION (figure 5). Although in patients with ON we hypothesised that expansion of the optic nerve sheath would be present due to the swelling of the optic nerve in the acute phase of the inflammation and consequent interruption of communication between the subarachnoid space of the optic nerve and chiasmal cistern,[2 26] no statistically significant differences were observed between the groups. The absence of sensitivity of optic nerve expansion seen in our study could be related to the time-to-imaging after symptom onset or beginning of therapy. As expected, most of the patients with ON showed an enhancement of the optic nerve that was statistically significantly different from that of patients with AION, in line with the results of Rizzo *et al* and Adesina *et al*.[4 14 15]

Although restricted diffusion is seen in many patients with acute ON[5 6 27–30] as well as in patients with AION and posterior ION (PION),[17 18 27 31–35] our results showed it had no significant added value in differentiating ON from AION, in keeping with the results of Adesina *et al*,[15] whereas our AION group included patients with arteritic AION. One explanation for our results could be that diffusion restriction was present in both groups; the MRI examinations assessed had no dedicated thin-section DWI sequences of the orbits; or, at the time of scanning, the signal alterations in the DWI were already reversed.

Recent advances in imaging have made the differentiation between AION and N-AION possible[36] by depicting the involvement, that is, the vessel-wall enhancement of the ophthalmic artery as well as of the posterior ciliary arteries. However, the analysis of these imaging techniques is outside the scope of this article, as our aim was to identify the MRI biomarkers that will make the differentiation between ON and AION (either arteritic or non-arteritic) possible.

The main limitations of our study are its retrospective character and the lack of a uniform MRI protocol for the orbits. The MRI scans analysed in our cohort were acquired between 2010 and 2018. Recently, new DWI techniques that complement the MR sequence spectrum have become available. These include readout segmentation echo planar imaging and reduced field-of-view DWI, which enable better visualisation of small brain structures. Due to the long timeframe of our study, these sequences were not always available. We are aware that it would be interesting to compare MRI results of patients with PION with those of patients with AION and ON, but that such a comparison was beyond the scope of the present study and would be worth investigating in the future. Finally, the relatively small number of patients could have limited the statistical power of this study.

CONCLUSION

Our findings suggest that in patients with vision loss where the clinical diagnosis is equivocal and MRI is being acquired, neuroradiologists should focus on the imaging findings that aid in the diagnosis and differentiation between ON and AION. The most important of these are the distribution of cerebral T2-hyperintensities, the presence and localisation of enhancement of the optic nerve and the 'central bright spot' sign. DWI added no useful information to the diagnosis of AION and/or ON. A prospective study of atypical clinical opthalmological cases should be considered to validate these observations.

Contributors VDP: collected the data, performed imaging and data analysis, co-wrote the manuscript draft. DB: collected the data, performed statistical analysis, co-wrote the manuscript draft. RH: collected clinical data. RV: contributed imaging data and analysis. AW: contributed imaging data. MA: conceived and designed the analysis and corrected the final manuscript. FW: conceived and designed the analysis, supervised the study, reviewed and corrected the final manuscript. Responsible for the ethical statement.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not required.

Ethics approval The study was performed in accordance with the principles of the Declaration of Helsinki and was approved by the local ethics committee (Kantonale Ethikkommission Bern, KEK, Basec PB 2019-02253).

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplemental information.

ORCID iDs
Dominik Brügger http://orcid.org/0000-0001-9837-6652
Mathias Abegg http://orcid.org/0000-0002-5150-9320
Franca Wagner http://orcid.org/0000-0001-7502-4814

REFERENCES

1 Biousse V, Newman NJ. Ischemic optic neuropathies. *N Engl J Med* 2015;372:2428–36.
2 Becker M, Masterson K, Delavelle J, *et al*. Imaging of the optic nerve. *Eur J Radiol* 2010;74:299–313.
3 Hayreh SS. Ischemic optic neuropathy. *Prog Retin Eye Res* 2009;28:34–62.
4 Hoorbakht H, Bagherkashi F. Optic neuritis, its differential diagnosis and management. *Open Ophthalmol J* 2012;6:65–72.
5 Balashov KE, Aung LL, Dhib-Jalbut S, *et al*. Acute multiple sclerosis lesion: conversion of restricted diffusion due to vasogenic edema. *J Neuroimaging* 2011;21:202–4.
6 Fatima Z, Motosugi U, Muhi A, *et al*. Diffusion-Weighted imaging in optic neuritis. *Can Assoc Radiol J* 2013;64:51–5.
7 Mathews MK. Nonarteritic anterior ischemic optic neuropathy. *Curr Opin Ophthalmol* 2005;16:341–5.
8 Kim DH, Jun JS, Kim R. Measurement of the optic nerve sheath diameter with magnetic resonance imaging and its association with eyeball diameter in healthy adults. *J Clin Neurol* 2018;14:345–50.
9 Wardlaw JM, Smith EE, Biessels GJ, *et al*. Neuroimaging standards for research into small vessel disease and its contribution to ageing and neurodegeneration. *Lancet Neurol* 2013;12:822–38.
10 Venables WN, Ripley BD. *Modern applied statistics with S*. 4th edn. New York: Springer, 2002.
11 Azen R, Traxel N. Using dominance analysis to determine predictor importance in logistic regression. *J Educ Behav Stat* 2009;34:319–47.
12 McFadden D. Conditional logit analysis of qualitative choice behaviour. In: *Frontiers in Econometrics*. New York, USA: Academic Press New York, 1973: 105–42.
13 FCS, Claudio BN. Dominance analysis, 2020. Available: https://cran.r-project.org/package=dominanceanalysis
14 Rizzo JF, Andreoli CM, Rabinov JD. Use of magnetic resonance imaging to differentiate optic neuritis and nonarteritic anterior ischemic optic neuropathy. *Ophthalmology* 2002;109:1679–84.
15 Adesina O-OO, Scott McNally J, Salzman KL, *et al*. Diffusion-weighted imaging and Post-contrast enhancement in differentiating optic neuritis and Non-arteritic anterior optic neuropathy. *Neuroophthalmology* 2018;42:90–8.
16 Remond P, Attyé A, Lecler A, *et al*. The central bright spot sign: a potential new MR imaging sign for the early diagnosis of anterior ischemic optic neuropathy due to giant cell arteritis. *AJNR Am J Neuroradiol* 2017;38:1411–5.
17 Al-Shafai LS, Mikulis DJ, Arterial O. Diffusion MR imaging in a case of acute ischemic optic neuropathy. *AJNR Am J Neuroradiol* 2006;27:255.
18 Al-Zubidi N, Stevens S, Fung SH, *et al*. Diffusion-weighted imaging in posterior ischemic optic neuropathy. *Can J Ophthalmol* 2014;49:e21–5.
19 Harrar DB, Solomon J, Shah AS, *et al*. Diffusion-weighted imaging changes in a child with posterior ischemic optic neuropathy. *Pediatr Neurol* 2018;84:49–52.
20 Mathur S, Karimi A, Mafee MF. Acute optic nerve infarction demonstrated by diffusion-weighted imaging in a case of rhinocerebral mucormycosis. *AJNR Am J Neuroradiol* 2007;28:489–90.
21 Quddus A, Lawlor M, Siddiqui A, *et al*. Using diffusion-weighted magnetic resonance imaging to confirm a diagnosis of posterior ischaemic optic neuropathy: two case reports and literature review. *Neuroophthalmology* 2015;39:161–5.
22 Kim MS, Jeong HY, Cho KH, *et al*. Nonarteritic anterior ischemic optic neuropathy is associated with cerebral small vessel disease. *PLoS One* 2019;14:e0225322.
23 Arnold AC. Pathogenesis of nonarteritic anterior ischemic optic neuropathy. *J Neuroophthalmol* 2003;23:157–63.
24 Wardlaw JM, Smith C, Dichgans M. Mechanisms of sporadic cerebral small vessel disease: insights from neuroimaging. *Lancet Neurol* 2013;12:483–97.
25 Miller NR, Arnold AC. Current concepts in the diagnosis, pathogenesis and management of nonarteritic anterior ischaemic optic neuropathy. *Eye* 2015;29:65–79.

26 Hickman SJ, Wheeler-Kingshott CAM, Jones SJ, *et al*. Optic nerve diffusion measurement from diffusion-weighted imaging in optic neuritis. *AJNR Am J Neuroradiol* 2005;26:951–7.
27 Lu P, Tian G, Liu X, *et al*. Differentiating neuromyelitis optica-related and multiple sclerosis-related acute optic neuritis using conventional magnetic resonance imaging combined with readout-segmented echo-planar diffusion-weighted imaging. *J Comput Assist Tomogr* 2018;42:502–9.
28 Lukewich MK, Micieli JA. Diffusion-Weighted imaging hyperintensity and low apparent diffusion coefficient of the optic nerve in myelin oligodendrocyte glycoprotein-IgG optic neuritis. *Can J Ophthalmol* 2020;55:e39–41.
29 Seeger A, Schulze M, Schuettauf F, *et al*. Advanced diffusion-weighted imaging in patients with optic neuritis deficit - value of reduced field of view DWI and readout-segmented DWI. *Neuroradiol J* 2018;31:126–32.
30 Wan H, He H, Zhang F, *et al*. Diffusion-weighted imaging helps differentiate multiple sclerosis and neuromyelitis optica-related acute optic neuritis. *J Magn Reson Imaging* 2017;45:1780–5.
31 Cauquil C, Souillard-Scemama R, Labetoulle M, *et al*. Diffusion MRI and tensor tractography in ischemic optic neuropathy. *Acta Neurol Belg* 2012;112:209–11.
32 Däubler B, Kamli A, Runge V. Readout-segmented diffusion-weighted imaging in a critical anatomic area – diagnosing posterior ischemic optic neuropathy (pion). *Fortschr Röntgenstr* 2014;186:1151–2.
33 Kilani R, Marshall L, Koch S, *et al*. DWI findings of optic nerve ischemia in the setting of central retinal artery occlusion. *J Neuroimaging* 2013;23:108–10.
34 Bender B, Heine C, Danz S, *et al*. Diffusion restriction of the optic nerve in patients with acute visual deficit. *J Magn Reson Imaging* 2014;40:334–40.
35 Finelli PF. Optic nerve enhancement and restricted diffusion in postoperative visual loss. *Neuroophthalmology* 2019;43:337–9.
36 Geiger J, Ness T, Uhl M, *et al*. Involvement of the ophthalmic artery in giant cell arteritis visualized by 3T MRI. *Rheumatology* 2009;48:537–41.